

PCR Product Analysis by Agarose Gel Electrophoresis

Laboratory Report — DNA Fragment Sizing by Standard Curve

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Technique category: Molecular Biology

1. Objective

To confirm PCR product identity and size using agarose gel electrophoresis, by fitting a DNA ladder standard curve and estimating product size from measured migration distance.

2. Introduction

After PCR amplification, agarose gel electrophoresis is used to confirm that the expected product was generated at the expected size. DNA fragments, being negatively charged, migrate toward the anode through the agarose gel matrix under an electric field; smaller fragments migrate further than larger fragments over a fixed run time. Running a DNA ladder of known fragment sizes alongside PCR products allows the size of each product band to be estimated by comparison, using the same log-linear relationship between fragment size and migration distance applied in SDS-PAGE protein sizing.

3. Materials

- PCR products and a DNA ladder (100 bp–2000 bp range)
- Agarose powder and TAE/TBE running buffer
- DNA loading dye and DNA-safe stain (e.g. SYBR Safe)
- Horizontal gel electrophoresis apparatus and power supply
- Gel imaging system (UV or blue-light transilluminator)

4. Method

- Prepared a 1.5% agarose gel with DNA-safe stain.
- Loaded PCR products (with loading dye) and DNA ladder into wells.
- Ran the gel at constant voltage (~100V) until adequate separation was achieved.
- Imaged the gel under UV/blue light and measured migration distances.
- Fitted a standard curve from ladder migration distances and estimated PCR product sizes.

5. Results

A standard curve fitted from twelve DNA ladder bands (100–2000 bp) gave **$\log_{10}(\text{bp}) = -0.0170 \times \text{distance} + 3.397$** (R-squared = 0.998), indicating an excellent linear fit. Two PCR products were assessed against this curve.

Product	Migration (mm)	Estimated Size (bp)	Expected (bp)	Result
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PCR_Product_1	58.0	258	230	Match
PCR_Product_2	75.0	133	230	Mismatch

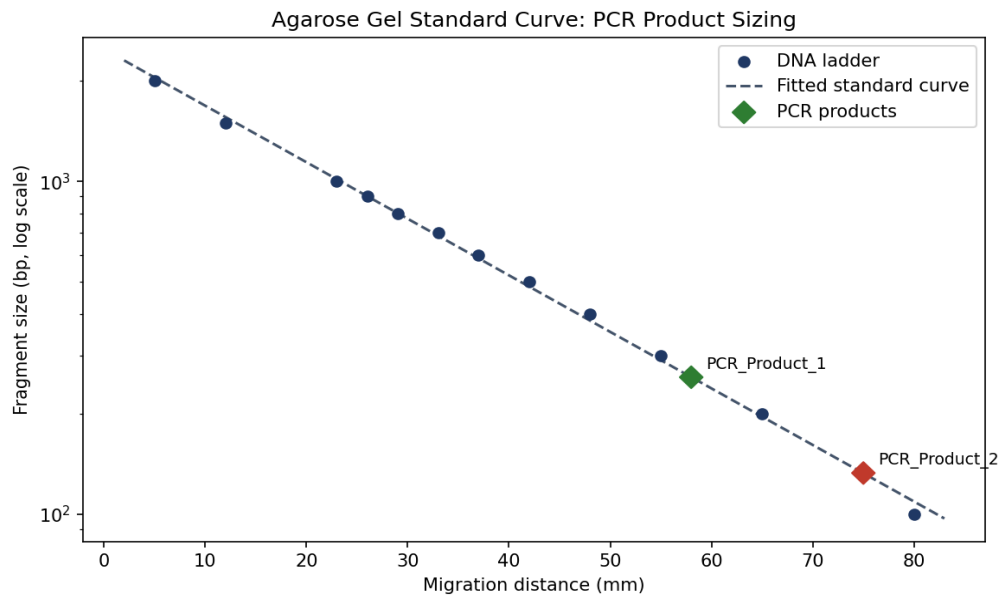


Figure 1. Agarose gel standard curve (log scale) with PCR products plotted against it; green = match, red = mismatch.

6. Discussion

PCR_Product_1's estimated size (258 bp) fell within the tolerance range of the expected 230 bp amplicon, confirming successful, specific amplification consistent with the primer design. PCR_Product_2's estimated size (133 bp) was substantially smaller than expected, indicating the band observed was not the intended target — a result consistent with non-specific amplification, primer dimer formation, or primer mis-annealing to an off-target site. In practice, this result would prompt troubleshooting steps such as re-checking primer specificity (e.g. via BLAST against the template), adjusting annealing temperature, or redesigning primers before repeating the reaction.

7. Conclusion

This exercise demonstrated the complete post-PCR confirmation workflow — from gel electrophoresis through standard-curve-based size estimation — and illustrated how gel-based sizing can distinguish successful amplification from non-specific or failed reactions, complementing the in-silico PCR simulation approach used earlier in this portfolio to predict expected amplicon size from primer positions.